EÖTVÖS LORÁND UNIVERSITY
FACULTY OF INFORMATICS

Thesis Topic Registration Form

Student's Data:
Student's Name: Chobanova Ulkar
Student's Neptun code: WQ1C71

Educational Information:
Training programme: Computer Science BSc

I have an internal supervisor

Internal Supervisor's Name: *Gregory Reynolds Morse*
Supervisor's Home Institution: ***Department of Programming Languages and Compilers***
Address of Supervisor's Home Institution: ***1117, Budapest, Pázmány Péter sétány 1/C.***
Supervisor's Position and Degree: *Lecturer and MSc in Computer Science*

Thesis Title: AI Cookbook with Ingredient-Based Search

Topic of the Thesis:
(Upon consulting with your supervisor, give a 150-300-word-long synopsis os your planned thesis.)

Problem Statement:
People often face challenges in deciding what to cook with available ingredients while considering dietary preferences or allergies. This project proposes an AI-powered cookbook that generates personalized recipes from a list of ingredients. The system will employ OCR to extract and structure nutritional information from food labels and provide structured recipe outputs with basic nutritional insights, enhancing usability and supporting informed cooking decisions.

Proposed Complex Features:
1. OCR-Based Nutrition Label Processing
Users can upload images of food labels. Text is extracted using an OCR framework (e.g., Google Vision API) and refined into structured JSON format by an LLM for consistent interpretation of nutritional values.
2. Structured Recipe Generation and Personalization
The application generates clear, structured recipes with sections for ingredients, preparation steps, and nutritional details using retrieval-augmented generation (RAG). Recipe suggestions are further personalized based on user preferences such as dietary restrictions or allergens.
3. Data Persistence and Backend Integration
The system will include a lightweight backend to handle recipe generation requests and manage user data. A database will ensure data persistence, allowing storage of user preferences, generated recipes, and interaction history for improved continuity and usability.

Budapest, 2025. 10. 13.